

PROJECT 1: CUSTOMER CHURN (Problems + Code 'Python')

http://localhost:8888/notebooks/Desktop/Telco-Customer-Churn.ipynb 110%

Jupyter Telco-Customer-Churn Last Checkpoint: 6 days ago Trusted

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JupyterLab Python 3 (ipykernel)

```
[5]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
from sqlalchemy import create_engine

[6]: engine = create_engine('mysql+mysqlconnector://root:2000@localhost/telcocustomerchurn')

[7]: query = 'select * from churndata'

[23]: df = pd.read_sql(query, engine).head(3)
df
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No

3 rows × 21 columns

```
: #1. Problem: Which customers churn the most?
df['Churn'].value_counts(normalize=True) * 100
```

```
Churn
No    66.666667
Yes   33.333333
Name: proportion, dtype: float64
```

#2. Problem: Which contract type has highest churn?

```
df.groupby('Contract')['Churn'].value_counts(normalize=True)*100
```

```
Contract      Churn
Month-to-month No    50.0
                Yes   50.0
One year      No   100.0
Name: proportion, dtype: float64
```

#3. Problem: Does MonthlyCharges affect churn?

```
df.groupby('Churn')['MonthlyCharges'].mean()
```

```
Churn
No    43.40
Yes   53.85
Name: MonthlyCharges, dtype: float64
```

#4. Problem: Which payment method causes churn?

```
df.groupby('PaymentMethod')['Churn'].value_counts(normalize=True)*100
```

PaymentMethod	Churn	
Electronic check	No	100.0
Mailed check	No	50.0
	Yes	50.0

Name: proportion, dtype: float64

#5. Problem: Do support services reduce churn?

```
df.groupby('TechSupport')['Churn'].value_counts(normalize=True)*100
```

TechSupport	Churn	
No	No	66.666667
	Yes	33.333333

Name: proportion, dtype: float64